

Quiz 10: Blood

Review of Session 10. ~15 minutes.

Section A: Typing

1. Type A blood has which antigen on the cells, and which antibody in the plasma?
2. Fill in the type for each agglutination result:

Anti-A	Anti-B	Anti-D	Type?
clump	—	clump	
—	clump	—	
clump	clump	clump	
—	—	clump	
—	—	—	

3. Name the universal red-cell donor and the universal recipient.
4. List the ABO/Rh blood types from most common to rarest.

Section B: Genetics

1. Which alleles are codominant? Which is recessive?
2. Can two type-O parents have a type-A child?
3. Can a type-AB parent have a type-O child?
4. A type-A parent and a type-B parent — what blood types can their child have?

Section C: Chemistry & tests

1. What protein in red blood cells carries oxygen?
2. What happens to blood, chemically, as it sits exposed to air?
3. Name two presumptive tests for blood and what a positive result looks like.
4. True or false: cyanoacrylate (superglue) fuming is a test for blood.
5. Does a matching blood type prove a suspect was at the scene?

Answer Key

Section A

1. A antigen on the cells; anti-B antibody in the plasma.
2. A+, B–, AB+, O+, O–.
3. Universal donor: O–. Universal recipient: AB+.
4. O+, A+, B+, O–, A–, AB+, B–, AB–.

Section B

1. A and B are codominant; O is recessive.
2. No — neither parent has an A or B allele to pass on.
3. No — an AB parent always passes an A or a B.
4. A, B, AB, or O (all four possible, if both parents carry an O allele).

Section C

1. Hemoglobin.
2. Oxygen binds the hemoglobin (brighter red); it clots within minutes; then the hemoglobin oxidizes and the stain turns reddish-brown to brown over hours.
3. Kastle-Meyer (turns bright pink); luminol (glows blue in the dark). (Also Hemastix.)
4. False — it develops fingerprints, not blood.
5. No — blood type is class evidence; it narrows the field but doesn't identify one person (DNA does).